# Dong Hwi Kim

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## Summary

I study 3D Gaussian Splatting for mapping and robot learning. My interests also include 3D vision, perception, control, robotics, and autonomous driving.

## Education

2021 – present B.S. candidate at **Seoul National University**
– Double major: Mechanical Engineering, Electrical & Computer Engineering

## Research Experience

**AIDAS Lab, Seoul National University** Jun. 2026 – present
*Undergraduate Student Researcher* Advisor: Prof. Jaeyoung Do

- Develop the 3D Gaussian Splatting augmentation module for a collaborative robot-learning dataset project.
- Align reconstructed FR3 scenes and replay robot motion using forward kinematics. Render demonstrations from calibrated viewpoints.

**RPM Robotics Lab, Seoul National University** Feb. 2026 – present
*Undergraduate Student Researcher* Advisor: Prof. Ayoung Kim

- Investigate 3D Gaussian Splatting for SLAM and geometrically consistent scene mapping.
- **First-author manuscript in preparation.**

**Visual & Geometric Intelligence Lab, SNU** Sep. 2025 – Jan. 2026
*Undergraduate Student Researcher* Advisor: Prof. Jaesik Park

- Studied vision-language models and multimodal visual understanding.

**Multiscale Biomedical Engineering Lab, SNU** Jan. 2025 – Mar. 2025
*Undergraduate Student Researcher* Advisor: Prof. NooLi Jeon

- Worked on deep learning-based medical image reconstruction.

## Awards and Honors

Engineering competition awards below are team awards.

**2nd Prize**, SNU AI Robot Challenge Jul. 2026

**3rd Prize**, Wind Power Forecasting Challenge Dec. 2025
3rd out of 270 teams; BDA, EWP, and 60Hertz

**3rd Prize**, SNU FastMRI Challenge Sep. 2025

**Grand Prize**, HD Hyundai Mechatronics Competition Jul. 2024

**3rd Prize**, Korean Mathematical Olympiad (KMO) Jun. 2019
Korean Mathematical Society [Certificate]

## Projects

**Autonomous Object-Collecting Robot** 2026
Project | GitHub
Contributed to perception, planning, data collection, hardware, and system integration. Our team combined dual-camera segmentation, geometric target selection, and LiDAR/IMU-based motion control. The robot placed **2nd** in the SNU AI Robot Challenge, scoring **140/200** in the final.

**Multimodal Scene Reordering** 2026
Project | GitHub
Team GeekSeek adapted Qwen3.5-27B with LoRA to order four shuffled frames. The submission scores valid permutations and aggregates predictions across input orderings. Our team finished **8th overall** in the SNU AI Challenge.

**Accelerated MRI Reconstruction** 2025
Project | GitHub
Worked with Team 2rm1000 on reconstruction from undersampled k-space. Our pipeline combines Feature/FI VarNet, data consistency, anatomy- and acceleration-specific models, and ensembling. The team placed **3rd** in the SNU FastMRI Challenge at 4×/8× acceleration.

**Physics-Guided Wind Power Forecasting** 2025
Project
Designed model architecture and optimized hyperparameters in Team Zephyr Analytics. We combined a physical turbine power curve with LightGBM residual learning and a weighted ensemble. Our team placed **3rd out of 270 teams**, reporting **19.34% NMAE**.

**Automated Filming Device** 2024
Project | GitHub
Built a hands-free filming prototype with a team. The Raspberry Pi system combines face detection, proportional pan–tilt servo control, and infrared remote input. The team received the **Grand Prize** in the HD Hyundai Mechatronics Competition.

## Work Experience

**ROK–U.S. Combined Forces Command** Sep. 2022 – Mar. 2024
Korean–English interpreter in a combined military environment.
Honors (2024): Joint Service Achievement Medal, U.S. Department of Defense; Deputy Commander's Commendation and Major General's Commendation, ROK–U.S. Combined Forces Command.

## Skills

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|---|---|
| Programming & Tools | Python, C/C++, PyTorch, Git, LightGBM, Optuna |
| Robotics & Vision | 3D Gaussian Splatting, robot kinematics, motion planning, visual servo control, YOLO segmentation |
| Languages | Korean (native), English (fluent), Chinese (conversational) |